Do Visual-Selection Operations Track Technical Quality? A Controlled Audit for Multi-Video Summarization

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Abstract

Multi-video summarization selects representatives from redundant content, and visual-selection operations developed for appearance or representativeness are reused to rank those representatives by technical quality without validation for that role. We test them in that role. The study combines source-level provenance, controlled degradation, external opinion-score evaluation, mechanistic analysis and representative substitution in a fixed modular testbed. We evaluate 15 literature-derived operations from 11 source works on 800 original–transformed comparisons drawn from 20 broadcast segments and on three public image-quality databases, alongside attribute configurations and dedicated no-reference quality models. Each of the 15 literature-derived scalar operations reverses on at least one of eight transformation types, and 13 reverse under white noise; nine keep a reversal under the most conservative unit of analysis, with one-sided exact tests and Holm correction. None meets both external screening conditions, nor do 19 attribute configurations, while 3 of 10 dedicated models do. A deep representativeness proxy, distance to the cluster centre, prefers the original in 74.1% of pairs, below every dedicated model, and still in 64.8% when the cluster is drawn from other topics: it mostly reads atypicality in the embedding. Over 42 redundancy clusters, 21 scoring implementations produce 18 distinct representative-selection sets. One audited operation factorizes exactly into a descriptor count and a mean rarity weight; replacing the score by the count leaves all 42 choices unchanged, so the operation reads local detail as quality. Implementation choice is consequential, and a reused operation should be validated against the requirement it is meant to serve.

Keywords: Multi-video summarization, No-reference image quality assessment, Content selection criteria, Benchmark evaluation, Broadcast news video

1 Introduction

Multi-video summarization selects content that represents shared and complementary information across source videos [3, 33, 41, 53, 59]. In a segment-based pipeline, redundancy detection groups candidate segments, and a subsequent selection criterion chooses a representative from each group. Human-strategy-based summarization makes these content-selection criteria explicit [5, 6], providing a modular setting in which one selection component can be examined while the others remain fixed.

A criterion states the property sought; its implementation supplies the decision rule and the measurements used by that rule. Here, the criterion examined is the quality criterion, and the property it is taken to seek is technical quality: fidelity of the visual signal under impairments such as noise, blur and compression artefacts, as distinct from semantic relevance, coverage, diversity and the usefulness of a whole summary. Operations developed for those purposes are reused to rank candidates by quality without being validated for that role. Tested in that role, the evaluated operations fail systematically, and which of them is used changes what the pipeline selects.

The visual-selection literature uses colourfulness, contrast, sharpness, entropy, saliency and local-feature statistics in several roles, and names them as quality: an average of five attributes is called a *quality score* [51], a weighted sum of sharpness and brightness evaluates frames on what its authors call *perceptual quality* [30], four features feed a *photo-quality* classifier [29], and one source names technical fidelity outright and implements it with BRISQUE [60]. Others enter as terms of multi-objective scores or as discard filters [8, 19, 23, 27, 44, 61]. Each source's evidence concerns the property or system it evaluates: an attribute may discriminate photographic styles, or a filter may remove blurred frames [20, 29]. Among the sources verified here, none validates the isolated reuse as a technical-fidelity score. This is a question of *construct validity* [13]: whether an operational measure represents the property attributed to it.

The question addressed here is whether the evaluated implementations remain suitable when used as standalone technical-quality scores. Summary-level benchmarks cannot answer it: F-scores on SumMe [19] and TVSum [50] evaluate a whole summary, and randomly generated summaries reach scores comparable to the state of the art [40], so a benchmark score does not isolate whether one criterion measures its property. The audit therefore works at the level of the component. Dedicated no-reference image quality assessment (NR-IQA) models provide relevant comparators because recordings from different broadcasters do not share an undistorted reference [1, 28, 36, 37, 65]. Their suitability is tested here on the intended data and decision.

We investigate the degradation response, external opinion-score association and selection consequences of these reuses, over fifteen operations reimplemented from eleven source works plus the ITU-T spatial information index, on 800 original–transformed comparisons drawn from 20 broadcast segments, and on three public

image-quality databases with synthetic (KADID-10k) and authentic (KonIQ-10k, LIVE In the Wild) distortions. The objective is to establish which of these operations respond to a technical-quality requirement, where they fail, and how much the choice among them changes what a pipeline selects. The study contributes (i) a source-linked inventory of operations and decision roles, each fixed by a verbatim excerpt; (ii) a common protocol connecting controlled perturbations and external opinion databases to component substitution; (iii) an exact decomposition and empirical diagnosis of the TF-IDF-weighted SIFT+BoVW score [5]; and (iv) a controlled measurement of representative-selection sensitivity within a fixed modular testbed. HSMVideo-Summ [18] serves as the modular testbed, and its own visual-word operation [5] is audited on the same footing as the others.

2 Related work

2.1 Visual selection in multi-video summarization

Extractive approaches select existing frames or segments, whereas abstractive approaches produce derived representations [3, 53]. Multi-video methods additionally address cross-source redundancy and diversity: Video-MMR formalized this setting [33], sparse and collaborative formulations select complementary content [41, 42], and graph-based methods represent inter-video relations [59]. Concurrent recordings of the same event provide another established setting for representative selection [4, 12].

Human-strategy-based summarization separates content-selection criteria into addressable stages [5, 6]. This modularity supports controlled substitution, while the evaluation analysis in [40] motivates holding segmentation and other protocol choices fixed. Image-quality assessment has already been used for video summarization: VSQUAL employed objective measures in shot-boundary detection and keyframe extraction [11], and a static-summary method combined representativeness with visual penalties [27]. What has not been examined on the selection side is the component-level comparison, source-role accounting and measurement of selection consequences reported here.

Table 1 organizes the implemented measures into four classes; the operations and their attribution to source works rest on the source-checked inventory of Section 3.1.

The documented roles are heterogeneous. Two of the source works combine attributes into frame scores [30, 51]. Other works place visual terms inside objectives that also address attention, representativeness, coverage or diversity [19, 23, 27, 44, 60, 61]. TriPSS uses visual signals in filtering [8]. One source introduced photographic-classification features [29], whereas another validated a colourfulness index [20].

2.2 No-reference quality assessment and transfer

Image quality assessment (IQA) predicts perceived degradation. Full-reference measures compare an image with an undistorted original [56, 63]; NR-IQA instead uses the test image alone. The latter matches representative selection across broadcasters, where no common original is available. The evaluated candidates include

Table 1 Four classes of scoring method for the quality criterion. *Ops*: operations instantiated in this testbed. *Papers*: of the 85 summarization papers in the archive, those in which a term sweep finds the technique mentioned, not its use as a quality score (triage, not confirmed adoption; Table S31). *Sources*: works whose operations are audited (Table 5). SIFT: scale-invariant feature transform; BoVW: bag of visual words; TF-IDF: term frequency–inverse document frequency.

Class	Ops	Papers	Sources	Example here
Low-level pixel heuristics	16	50	11 works (Table 5)	Laplacian variance, colourfulness
Visual-word counting	3	39	[5]	SIFT+BoVW with TF-IDF
Dedicated NR-IQA models	10	1	[60]	BRISQUE, NIQE, LIQE, MUSIQ
Deep representativeness proxies	3	23	[27]	Distance to cluster centroid

natural-scene-statistics models, BRISQUE and NIQE [36, 37]; opinion-supervised convolutional neural network (CNN) and attention models, DBCNN, HyperIQA, MUSIQ and TOPIQ [10, 28, 52, 64]; and vision–language models based on Contrastive Language–Image Pre-training (CLIP) [45]. These include CLIP-IQA and CLIP-IQA+ [54], LIQE [65] and QualiCLIP [1]. Q-Align maps multimodal-model outputs to discrete quality levels [58].

Architecture and exposure to opinion data both vary across released checkpoints. An association measured on a training database is not evidence of transfer to that database, and strong in-database performance need not persist in a new domain [9, 43]. We therefore record opinion-data exposure and apply the external-transfer condition only to unseen databases. KADID-10k provides known synthetic distortions and levels [34], while KonIQ-10k and LIVE In the Wild provide authentic distortions with human opinion scores [16, 21].

Image-wise scoring cannot observe all temporal impairments, motivating a separate video-model assessment of selected outputs [24, 55, 57]. Correlation also depends on the range and uncertainty of subjective scores [31]. Paired-comparison designs [7, 26, 39] complement database-wide association by asking whether an operation orders controlled alternatives in the intended direction.

2.3 The measurement gap

Scale-invariant feature transform (SIFT) descriptors, bag-of-visual-words (BoVW) representations and term frequency–inverse document frequency (TF-IDF) weighting encode feature occurrence and rarity for recognition or retrieval [14, 35, 47, 49]. Spatial information describes scene complexity [25, 46]. One of the works audited here applies exactly this combination to multi-video summarization: the audited proposal scores a segment by the sum of TF-IDF weights over a visual-word vocabulary built

from SIFT descriptors [5], and HSMVideoSumm is the released implementation of that pipeline [18]. That operation is evaluated here on the same footing as the others.

Because the intended task is part of how a quality measure is interpreted [32], our design links the source-assigned role to observed score behaviour and selection consequences, separating three questions: what an operation was designed to measure, what it measures under the present tests, and what changes when a pipeline uses it.

3 Audit design

The study combines provenance analysis with four complementary tests: controlled degradation, external opinion-score evaluation, mechanistic diagnosis and representative-selection sensitivity (Figure 1). The main scalar inventory contains fifteen directionally specified operations catalogued from eleven source works, plus the International Telecommunication Union's spatial-information index, which has no quality polarity and is excluded from directional screening. To these are added ten dedicated NR-IQA models, which form the complete-case comparator set.

Three other sets have distinct purposes. The nineteen attribute configurations test combinations of the implemented statistics; 21 scoring implementations, the fifteen audited operations and six further terms that enter only this comparison (Figure 2), support the application-stage comparison; and the SIFT+BoVW operation supports an internal decomposition. That score is not one of the sixteen and is screened by the same rule. These sets overlap, and Table S18 gives the coverage of each analysis for the sixteen audited operations.

Eligibility is a study-defined screening rule that combines response to synthetic degradation, association with human opinion on unseen databases, and above-chance agreement with a designated model in the broadcast domain; the first two are the external conditions.

3.1 Identifying the operations, and reproducing the published rules

The experimental inventory was selected before the principal scoring runs from operations associated with visual selection in the cited literature. Inclusion required an isolated-frame statistic and a declared direction for that statistic. Graph-based multi-video methods such as MvsGCN [59] score frames through learned representations and name no quality statistic that can be isolated per frame, so they fall outside this rule. Table S17 lists the operations that meet it, reimplemented from their sources. The inventory therefore holds three kinds of row: expressions the source specifies, implementations adapted at the input or the normalization, and *combinations aggregated here* from attributes a source names without stating how it combines them. The last kind enters because each attribute it averages satisfies the rule on its own; the aggregation is ours and is marked as such in Table S22. The coverage accounting for the archive sweep, the search patterns and the limits of that instrument are recorded in Table S30.

Table S16 states the objective and role recorded for each source, and the verbatim excerpt that fixes the validation evidence is in Tables S24 and S25. Standalone

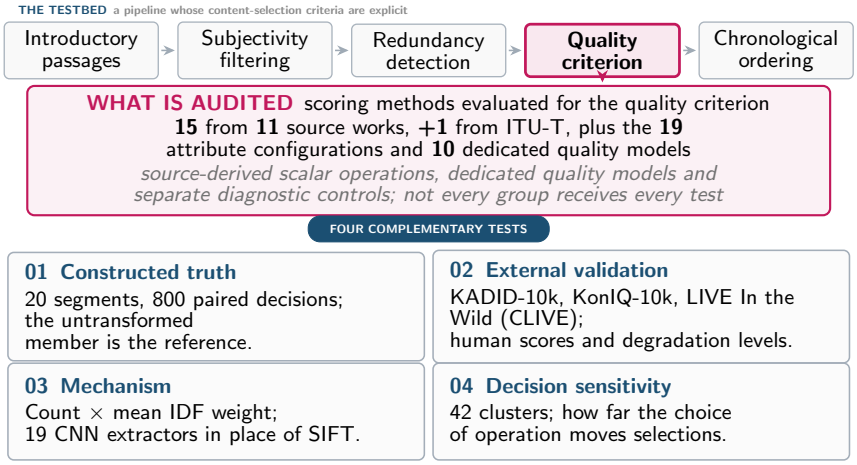


Fig. 1 Scope and evidence flow. **Top**: the content-selection criteria of a segment-based multi-video summarizer, with the quality criterion as the one under study. **Middle**: the scoring methods audited as implementations of that criterion, from the provenance audit. **Bottom**: the four tests. Each test reaches a different subset of the audited operations, recorded in Table S18. All other pipeline stages are held fixed (Section 3.2).

ordering is distinguished from classifier features, terms within a larger objective and threshold-based filtering. Validation evidence is coded by its object, attribute, module or summary, and reference, such as human ratings, summary evaluation or ablation. All eleven source works were read in full.

Implementations reproduce the specified expression where its inputs are available; task and interface adaptations are recorded separately in Table S22. In particular, saliency area uses a spectral-residual map [22] rather than the source's learned map; the rule of Ref. [30] uses normalization over the benchmark candidates, including original and transformed members; and the comparator of Ref. [29] is a Gaussian naive Bayes classifier trained on constructed-degradation labels with leave-one-segment-out validation. These variants evaluate adapted components.

Polarity is attached to the actual quantity extracted. In Ref. [27], raw pixel variance is transformed into a badness penalty before entering a minimized cost. The present statistic is the grey-level dispersion, implemented as the standard deviation rather than the variance the source states. That source names three badness terms, contrast, blur and skew, inside one multiplicative cost, and only the contrast term is audited here: the other two are not separable from the published expression as isolated-frame statistics, which is the same inclusion rule that admits the contrast term. A second polarity ambiguity appears in Ref. [60], which states that all five entity scores are normalized to $[0, 1]$ with higher meaning better while using BRISQUE, whose published direction is lower-is-better, without stating an inversion; the direction evaluated here is the one the scorer itself declares.

3.2 Data and units of analysis

KADID-10k contains 10,125 distorted images generated from 81 references, 25 distortion types and five levels [34]. Its released opinion-score field is named ‘dmos‘ but increases with quality. KonIQ-10k and LIVE In the Wild, also denoted CLIVE, supply authentic distortions and Mean Opinion Scores (MOS) [16, 21]. No dedicated NR-IQA model is retrained or fine-tuned. The fitted combination variants are identified separately.

HSMVideoSumm provides the modular broadcast-news testbed [5, 6, 18]. Its visual-word operation is one of the audited works of this study: the source proposes scoring a segment by the sum of TF-IDF weights over a $K = 300$ visual-word vocabulary [5], and the released code is the implementation these measurements run [18]. **It was chosen because it is modular, open and documented at the stage this audit targets**: the quality criterion is a replaceable component, so an operation can be swapped without touching the rest of the pipeline, which is what the substitution analyses require. It was also developed in our laboratory, which gave access to the released code and to the corpus it runs on. That source defines the criterion of the stage audited here as whether a segment *contains images of interest*, a relevance criterion, and the same study reports that aesthetic quality was not relevant to its participants. We reuse its modular pipeline to evaluate a separate requirement, technical fidelity, defined in this audit. Original redundancy groups and the remaining pipeline stages are retained.

The application uses the 42 multi-member clusters produced by the redundancy stage over 19 of the 21 topics, containing 93 segment slots and 83 distinct segments. Some segments occupy more than one cluster. Still-image scorers ordinarily receive frames sampled at one frame per second and use the segment mean

$$S(s) = \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(f_i), \tag{1}$$

where $\phi(f_i)$ is the frame score. The SIFT+BoVW counting operation instead uses its own keyframe selector and a vocabulary of $K = 300$ visual words.

Each implementation ranks candidates within a fixed cluster and selects the highest-ranked candidate after polarity alignment. The two treatments are reported separately and distinguish the effects of implementation and scoring. The complete-implementation comparison preserves each operation’s own sampling and aggregation. The matched-input control supplies both scorers with the same, at most ten, uniformly spaced frames and mean pooling. Only the matched-input control isolates scoring under a common input regime.

The independent unit follows the analysis: KADID reference image for synthetic-distortion sensitivity; source-aware topic block for the primary constructed-pair inference; and news topic for application inference. Segment-block and hierarchical alternatives quantify sensitivity to dependence assumptions. Individual frames, transformations and overlapping segment slots are not treated as independent replicates.

Table 2 No-reference comparators and reference. *Opinion training*: the database whose human scores trained the released checkpoint; *none* marks opinion-unaware models. The transfer condition is tested on every evaluated database (KADID-10k, KonIQ-10k, LIVE In the Wild) absent from that training; the LIVE database of the BRISQUE checkpoint is not among them, so all three are unseen for it. *Pol.*: declared polarity, fixed before evaluation; ↑, a higher score means better quality; ↓, a lower score means better quality. Constructs, released weights, endpoints and preprocessing, with the diagnostic controls and the 19-extractor ablation, are in Table S12.

Method	Opinion training	Pol.
Candidates, opinion-unaware		
NIQE	none	↓
CLIP-IQA	none	↑
QualiCLIP	none	↑
Candidates, opinion-trained		
BRISQUE	LIVE [48]	↓
DBCNN	KonIQ-10k	↑
HyperIQA	KonIQ-10k	↑
MUSIQ	KonIQ-10k	↑
TOPIQ	KonIQ-10k	↑
CLIP-IQA+	KonIQ-10k	↑
LIQE	KonIQ-10k	↑
Reference		
Q-Align	multiple databases	↑
Partial coverage, outside screening		
ARNIQA	KonIQ-10k	↑
TReS	KonIQ-10k	↑

3.3 Scorers, references and polarity

The complete-case comparator set contains ten no-reference image quality assessment (NR-IQA) models: BRISQUE, NIQE, DBCNN, HyperIQA, MUSIQ, TOPIQ, CLIP-IQA, CLIP-IQA+, LIQE and QualiCLIP. ARNIQA and TReS extend completed axes but do not enter complete-case screening [2, 17]; MANIQA was incomplete at the scoring freeze [62]. Released checkpoint, supervision, preprocessing and decision polarity identify the evaluated configuration in Table 2.

Diagnostic controls include keypoint density, spatial information, a second inverse-document-frequency formulation and deep-feature measurements. The deep controls use DINOv2 [38], specifically ViT-S/14 with registers [15] at half precision (FP16), averaging patch tokens into a 384-dimensional frame vector. They measure vector norm, the negated mean distance of frames to their segment centroid, and distance to the centroid of the operational cluster, which Ref. [27] adds to its quality terms as the distance of a frame from its cluster centre; here it is computed between segment vectors. On the constructed pairs the degraded copy joins the original's cluster and the

centroid of all members decides between the two. A separate ablation over 19 convolutional extractors replaces SIFT while preserving vocabulary construction, weighting and aggregation. Two counting-descriptor substitutions were also excluded. SURF is excluded for lack of a redistributable implementation, being patented and absent from the OpenCV build these measurements ran on. The second was excluded because exchanging it would require recalibrating the key-frame selector that runs before the score. Table S12 records configurations, endpoints and the build environment.

Q-Align, an architecturally distinct model at the inventory freeze, is the in-domain reference for the NR-IQA candidate set; LIQE is the reference for the literature-derived operations. The two groups are compared directly only in a common-reference analysis (Section 4.5).

DOVER [57] scores the selected outputs as model-based evidence of output quality, independently of eligibility; its protocol and results are in the supplement.

For a score q and target y, direction-aligned Spearman correlation is $\rho^* = s_m s_y \rho(q, y)$, where $s_m = +1$ for a higher-is-better score and -1 otherwise; s_y specifies the target's direction. All released opinion targets used here have $s_y = +1$. In contrast, larger distortion levels d denote more degradation, so sensitivity requires $s_y = -1$ and $\rho^*_{\text{level}} = -s_m \rho(q, d)$. This target-specific definition separates a polarity convention from a genuine reversal.

For the nineteen attribute configurations, an additional oracle-polarity diagnostic uses the absolute correlation separately on each database or axis. It gives a favourable retrospective orientation, since the sign is inferred from the test outcomes; conclusions about transfer rely on the declared directions.

3.4 Controlled degradation and constructed truth

A seeded draw without replacement selects twenty of the 83 broadcast segments, which come from thirteen of the 19 operational topics (Table S2); two of those labels share source videos and overlapping segments, so they are merged into 12 source-aware blocks. Eight deterministic transformations at five levels produce $20 \times 8 \times 5 = 800$ original–transformed comparisons per fully covered operation. The transformations are darkening, motion blur, Gaussian blur, H.264 recompression, downscale-and-restore, desaturation, white noise and JPEG compression.

The untransformed member is the fidelity reference the protocol defines. A sensitivity analysis excludes darkening and desaturation, which need not worsen appearance for every viewer. With W, L and T the pairs in which an operation prefers the untransformed member, the transformed one or neither, *strict* accuracy is $W/(W + L + T)$ and *decisive* accuracy $W/(W + L)$; the two are not interchangeable, so the main tables report strict accuracy, the per-cell tests decisive accuracy, and a descriptive inversion is $L > W$. Only four of the fifteen operations tie at all.

For each operation–transformation cell, exact sign-flip randomization uses source-aware block means and enumerates all 2^b sign assignments. The reported directional tests use the lower tail with Holm correction over fifteen operations and eight transformations, or 120 tests. The convention matters at this sample size: with 12 blocks the smallest attainable one-sided p is 2.44×10^{-4} against a first Holm threshold of 4.17×10^{-4}, so a two-sided convention would reject nothing. Segment-level results

provide an alternative dependence assumption. Table S27 reports cell-level tests and Table S26 the alternative intervals.

A second analysis correlates score with transformation level within each segment–type combination. On KADID, correlation across five levels is computed separately for each reference image and distortion type before averaging. This prevents the many derivatives of one reference from inflating the number of independent source images.

3.5 Screening and application inference

Scoring methods are evaluated against three eligibility conditions, which admit methods without ordering them. The sensitivity condition uses KADID only. It requires mean direction-aligned Spearman Rank Correlation Coefficient (SRCC) of at least .50, with at least 15 of the 25 distortion types exceeding .50 after reference-wise averaging. The eight-type broadcast experiment is complementary and does not determine this condition. The external-transfer condition requires database-wide direction-aligned SRCC of at least .65 on every database absent from a checkpoint's opinion training. Threshold sensitivity is reported in Table S9.

The two aggregations of attribute configurations are author-defined. The z-score mean re-estimates a mean and a standard deviation per operation on each database, so its KonIQ-10k figure uses a normalization fitted on KonIQ-10k. The ridge is fitted on KADID-10k with the distortion level as target and a fixed penalty of one over standardized inputs, so its KADID figures are in-sample and its KonIQ-10k figure applies the KADID coefficients. Both are reported as diagnostic associations, and the in-sample ridge is an optimistic diagnostic that only shows whether a negative outcome survives the more favourable aggregation.

The in-domain condition requires above-chance agreement with the designated reference and a positive lower bound of its 95% topic-block bootstrap confidence interval. For cluster c with k_c candidates, independent uniform choices agree with probability $1/k_c$. Across the common 42 clusters, $\bar{p}_e = |C|^{-1} \sum_{c \in C} 1/k_c = .470$, and normalized agreement is $\kappa_{\text{uniform}} = (p_o - \bar{p}_e)/(1 - \bar{p}_e)$. This is a uniform-choice correction, not Cohen's kappa.

The principal inventory, scoring runs and caches were frozen on 23 August 2026; dependence-aware derived analyses were subsequently computed from those caches, and the deep-proxy controls were completed on 21 September, with the cluster-centre test on the constructed pairs registered and run on 27 September. Archived prediction statements refer only to their specified diagnostic tests.

Use of large language models.

Language-model assistants were used to help write the analysis and verification scripts, under the authors' supervision. Every quantitative statement reaches the manuscript through a macro emitted by a versioned script from a stored result file, and is checked by the consistency suite released with the package; the authors took every methodological decision and are responsible for the content. The full declaration, naming the assistants, is in the Statements and Declarations.

4 Results

4.1 Classes of scoring method, and what the testbed instantiates

Table 1 gives what the testbed instantiates in each class, and the screening of Section 4.5 covers the scalar operations that the isolated-image protocol admits. Of the three deep proxies implemented over the DINOv2 ViT-S/14 embedding, only the norm of the feature vector is an absolute per-frame score. Frame coherence needs several frames and the distance to the cluster centre, the quality term of Ref. [27], needs a candidate set, so both are *not applicable* to the external databases, which score isolated images. The feature-norm reading is a construction of ours used as a diagnostic: no audited source states it. All three appear in the last band of Table 3.

The class does not fail as a block. The norm averages 22.4 per cent and ranks the degraded copy above the original on 7 of the eight types: degrading a frame lengthens its representation, so reading vector length as quality is reversed. It fails outside the pairs as well: its lowest correlation on an unseen database is $-.074$, and its agreement with the reference on the operational clusters $-.258$. Frame coherence averages 66.2 per cent and inverts on 1 type. Its score is the negated mean distance of the frames to their own centroid, so a per-frame perturbation that scatters the representations, white noise among them, lowers it. Both readings were registered as opposing predictions before the measuring script was run. On the pipeline's own clusters frame coherence agrees with the reference at .326, a point estimate without the interval the in-domain condition requires.

Two candidates are equidistant from their own midpoint, so the cluster-centre distance cannot decide a pair alone. Placed in the original's operational cluster, with the centroid computed over all members, it prefers the original on 74.1 per cent of the 800 pairs (95% CI $[67.5, 80.2]$ by segment) and on no type falls below 56.0, under every dedicated model of Table 3, the lowest of which reaches 80.8. Most of that preference does not need the cluster: in a control added after this result, members drawn from other topics leave 64.8 per cent, and the real cluster adds 9.2 points, computed pair by pair ($[2.8, 16.4]$). The proxy mostly registers that a degraded copy moves away from typical broadcast content in the embedding. On the operational clusters it decides only in the 6 with three or more members, too few for the in-domain condition.

4.2 Responses to controlled degradation

All fifteen operations prefer the degraded copy on at least one type; thirteen do so under white noise, and five are wrong on all 100 noise pairs (Figure S3).

Under the one-sided, Holm-corrected randomization of Section 3.4, how many inversions survive depends on the nested unit. Over the 20 segments, 22 of the 35 cells below chance survive, in fourteen of the fifteen operations; over topics, the more conservative reading, 11 cells in nine operations. The topic unit is twelve source-aware blocks; keeping the two shared labels apart would leave 15 cells. What does not depend on the unit is the core: the five cells in which an operation is wrong on all one hundred pairs survive in every unit. The same test leaves seven of the ten published models

with no type below chance, and removing the two types aligned with an operation's own attribute leaves all fifteen still inverting. Table S27 lists the 35 cells with the verdict under each unit, and Table S26 recomputes every interval under the rival unit.

White noise concentrates the failures, and it is the only tested transformation whose median energy ratio exceeds one. Measured as the median ratio of Laplacian variance after to before, one of the eight types raises it: noise multiplies it by 6.9, every other type leaves it at or below 1.00 of the original, JPEG included at 0.95; Figure S2 shows the paired accuracy of the fifteen operations against the same eight types. Laplacian variance indicates local high-frequency content; these operations are statistics of local structure, and additive noise raises such statistics, so an operation reading more structure as more quality prefers the noisier frame. Across the fifteen that is a pattern, and Section 4.3 measures the chain end to end for the SIFT+BoVW score.

The same 800 pairs place the evaluated scalar scorers, the no-reference models and the three deep proxies on one scale (Table 3). SIFT+BoVW is not an outlier among the scorers this literature supplies: it is correct on 65.1% of the pairs, which sits at the median of the literature-derived operations (66.0%; Table S32). LIQE is correct on 99.5%, a paired difference of +34.4 points, with 95% CI $[+30.4, +39.4]$ over the 12 source-aware blocks, $[+29.8, +38.9]$ by segment block and $[+30.3, +39.1]$ under the separate-label grouping. Table S32 records the literature operations with their accuracies and the comparisons behind each.

Excluding darkening and desaturation leaves 600 comparisons and does not repair the SIFT+BoVW operation: its accuracy is 63.7%, whereas LIQE reaches 99.8% and Q-Align 100% (Table S11).

The average conceals the diagnostic result: SIFT+BoVW is at or above 88% on five types but selects the degraded copy in 92% of white-noise pairs and 94% of JPEG pairs. Section 4.3 examines what the descriptor count and the inverse-document-frequency weighting contribute to those reversals.

Q-Align is correct on 98.2% of these pairs (95% CI $[97.2, 99.1]$), which supports its technical-degradation competence on this broadcast content. The exact downstream DOVER technical branch (DOVER-T) output reaches 97.6% ($[96.1, 99.0]$), or 99.8% ($[99.5, 100]$) without darkening and desaturation. The registered prediction audit, including the two overlay predictions, is in Table S11.

4.3 Why a count of local detail reads as quality

The decomposition below separates the contribution of the descriptor count from that of the mean rarity weight.

$$\text{score}_i = \sum_v \text{tf}_{iv}\text{idf}_v = D_i \underbrace{\sum_v p_i(v)\text{idf}_v}_{\overline{\text{idf}}_i}. \tag{2}$$

Here D_i is the number of descriptors the extractor finds in segment i, tf_{iv} how many fall in visual word v, and $p_i(v)$ that count as a fraction of D_i, so the score is a descriptor count times a mean inverse-document weight, and which channel dominates is measured rather than assumed. The identity is algebraic and holds for any weights,

Table 3 **Strict** accuracy $W/(W + L + T)$, in per cent, on 800 unmodified–degraded broadcast pairs: 20 segments, eight types and five levels. *dark*: darkening; *motion*: motion blur; *blur*: Gaussian blur; *H.264*: video recompression; *rescale*: downscale and restore; *satur.*: desaturation; *noise*: white noise. Values below the 50% chance floor are **descriptive inversions** where $L > W$, in bold and in colour; which of them survive the corrected exact tests is reported in the text and in Table S27. No row ties, so strict and decisive accuracy coincide, except the cluster-centre distance, whose 15 pairs with the original in two disagreeing clusters count one half. The first band holds SIFT+BoVW and the three best-scoring literature operations. ARNIQA appears on this paired axis but stays outside the complete-case screening of Section 4.5. The last band holds the deep proxies of Section 4.1; for the cluster-centre distance the degraded copy joins the original's cluster.

Method	dark	motion	blur	H.264	rescale	satur.	noise	JPEG	mean
SIFT+BoVW	100	100	90	90	88	**39**	**8**	**6**	**65.1**
Mean rule [51]	100	100	100	100	100	98	**29**	69	**87.0**
Colourfulness [20]	100	100	100	100	100	100	**3**	79	**85.2**
Averaged aesthetics [19]	100	100	100	100	100	100	**3**	67	**83.8**
LIQE	97	100	100	100	100	100	99	100	**99.5**
TOPIQ	93	100	100	100	100	93	100	100	98.2
MUSIQ	93	100	100	100	100	86	100	100	97.4
QualiCLIP	99	100	95	100	96	76	96	71	91.6
ARNIQA	91	100	99	98	100	61	82	98	91.1
DBCNN	84	97	92	97	95	92	85	81	90.4
NIQE	100	100	98	97	98	**0**	100	96	86.1
CLIP-IQA+	76	99	79	100	91	87	84	64	85.0
HyperIQA	**47**	100	100	100	100	**30**	100	100	84.6
CLIP-IQA	58	100	83	100	92	91	55	76	81.9
BRISQUE	100	100	100	99	100	**0**	67	80	80.8
Feature-vector norm	**33**	**10**	**19**	**10**	**19**	60	**14**	**14**	**22.4**
Frame coherence	62	50	76	**44**	77	63	72	86	66.2
Cluster-centre distance	56	86	65	77	71	76	91	72	74.1

since $p_i(v) = \mathrm{tf}_{iv}/D_i$ by definition, whenever the aggregation is a sum over raw term frequencies and $D_i > 0$. Averaging frame scores leaves a count-per-frame factor, $(D_i/F)\,\overline{\mathrm{idf}}_i$ over F frames, and only dividing by the descriptor count removes it.

In the deployment regime the count channel dominates: $\rho(S_{\mathrm{dep}}, D) = +.995$, and on the 800 constructed pairs the signs of the score and count differences agree in 797, the three exceptions differing by at most four descriptors against a median count of 1,092. **Replacing the score by that count leaves every representative choice unchanged**, 42 of 42 on the application clusters, while replacing it by the number of key frames the selector retains preserves only 23; Table S33 reports the reduction. The degraded copy carries more descriptors in 92% of noise pairs and 95% of JPEG pairs, so noise and blocking raise the count channel of Eq. 2 while structure-removing distortions happen to move it in the direction of fidelity: the score is responsive, but not consistently to quality. The 19 convolutional extractors average 44.9% (CI $[41.0, 48.9]$) and remain dominated by their descriptor count, which refutes the registered prediction that fixed feature maps would force chance performance.

Exchanging the extractor therefore changes individual choices without making the ordering a better predictor of quality, and the normalization variants evaluated here did not repair the reading either. Figure S1 shows the two channels; the supplement

gives the exact relation between the two inverse-document-frequency formulations, which differ by an additive count term, and the per-extractor external correlations.

4.4 Literature-derived composites and source-specified rules

Nineteen attribute configurations combine the implemented statistics (Table S29). Twelve combine more than one term and seven are single terms, aggregated by a z-score mean or a ridge model defined here. None of the nineteen satisfies both external screening requirements under either aggregation. One z-score combination passes sensitivity with a within-reference mean of .524 on 17 of 25 types, above the requirement no individual operation met, but its KonIQ-10k correlation is $\rho^* = .061$ against a requirement of .65.

Several sources specify how their own terms combine, and those specifications were implemented over the same attribute vector so the comparison isolates the rule. In both sources that specify a complete module, the published rule scores below its own best component, and each contrast excludes the degradation type aligned with that module's own construction, so it runs on seven types rather than eight. For Ref. [44] the contrast excludes satur and uses 700 comparisons from 20 segments: 74.3 per cent against 83.1 for its best term, colourfulness, a paired contrast of 8.9 points with an interval of $[7.3, 10.3]$ that excludes zero. For Ref. [30] it excludes dark and uses 385 comparisons from the 11 of 20 segments carrying both inputs: 64.2 against 69.6, a contrast of 5.5 points with a segment-block interval of $[0.0, 10.6]$ that touches zero. The source-level synthesis of Section 4.7 reports the corresponding all-eight-type means instead, 77.5 and 68.6 per cent, which is why the two sets of numbers differ. The comparator is each module's best component, chosen after the results were seen. The adapted classifier of Ref. [29] reaches 69.9 per cent against 87.0 for the unweighted mean, as a task adaptation (Section 3.1).

In two sources the published expression carries the opposite sign to the property the surrounding text names for it: the contrast term of the objective of Ref. [44] is the difference between the standard deviations before and after histogram equalization, not a measure of contrast itself, and in a controlled sweep over 25 levels it is perfectly inversely associated, at -1.0000, with the contrast used to build the inputs. The result is established for the tested sweep. The brightness term of the module of Ref. [30] needs no such condition: it is described as penalizing over-exposure while the mean luminance it specifies is monotone in exposure and enters with a positive weight. Both statements concern the formulations as published; Table S21 gives the sweep.

4.5 External validation and screening

No evaluated operation clears either external eligibility condition. The best within-reference mean among the fifteen is .298 against a requirement of .50, and the best lowest-unseen correlation is .139 against .65; spatial information carries no verdict, reaching .218 only in an increasing direction its source does not declare. **Which database decides the failure is not fixed**: for twelve of the fifteen operations the worst case is on the synthetic KADID-10k and for the remaining three it is on the authentic KonIQ-10k, so the generalization axis is not carried by a single database.

Table 4 reports each group's best value per axis. The result is stable: across 10,000 resamples of the reference images, none of the fifteen changes verdict, and four operations are *negative* on the first requirement, their response running against the direction their own use assumes. **The fifteen screened literature operations all belong to the first class**, and SIFT+BoVW is the one screened member of the second: the isolated-image protocol admits scalar per-frame statistics, and that is what the summarization literature supplies. Table 1 shows the other side of the same asymmetry, that the class of dedicated models, with ten operations instantiated here, is matched to one paper in the archive. The narrow coverage of this screening therefore follows from what the field publishes.

Table 4 External screening by method class (Table 1). Each cell is the maximum over its class on that axis alone, so **the two maxima need not belong to the same member**. *Clears both external* counts the members passing sensitivity and generalization together; **the in-domain floor is a third condition, reported separately**. Spatial information declares no direction and carries no verdict; the heuristics row counts 16 computed and 15 evaluated, and the bag-of-visual-words row 3 computed and 1 evaluated: the other two, keypoint density and the 19-extractor ablation, are diagnostics rather than published scores. Deep proxies read context and fall outside the isolated-image protocol (Section 4.1). The two source-specified rules are not screened (Section 4.4). ⋆Diagnostic, not a published class: generalization is the KonIQ-10k association under a data-derived orientation with normalization re-estimated per database, which makes its zero conservative. Per-operation values are in the supplement.

Evaluated group	n	Best sens.	Best gen.	Clears both external
Low-level pixel heuristics	16	+.298	+.139	0 of 15 evaluated
Visual-word counting: SIFT+BoVW [5]	3	−.077	−.445	0 of 1 evaluated
Dedicated NR-IQA models	10	+.849	+.723	**3 of 10**
Deep representativeness proxies	3	—	—	not screened (Section 4.1)
Attribute configurations (diagnostic)⋆	19	+.524	+.601	0 of 19 (1 clears sens.)

Three dedicated models are eligible, LIQE, QualiCLIP and CLIP-IQA+, and the set is stable under the threshold variations of Table S9. Within-reference KADID sensitivity separates the SIFT+BoVW operation sharply: its mean is −.077 (95% CI [−.124, −.030]) with 5 of 25 types above .50, against .807 to .849 for the three eligible methods. On the in-domain floor, against LIQE, the reference the protocol assigns to the literature-derived operations, it agrees on 20 of 42 clusters ($\kappa_{\text{uniform}} = .011$), an interval that crosses chance. Several literature-derived operations meet this condition on unmodified broadcasts despite failing both external ones, and three lose that pass when one topic is removed. The sensitivity verdict is also conditional on the benchmark: applied to the constructed pairs, the same rule admits six of the fifteen, and none

meets it on both. The per-model floor intervals, the common-reference comparison, the regime term behind the KADID gap and the cost are in the supplement.

4.6 Effect of the implementation on content selection

Across the 42 operational clusters, 21 evaluated scoring implementations of the one criterion produce 18 distinct representative selections, with the decoded frames and the remaining criteria held fixed, and at least two disagree in 42 of the 42. Every candidate in every cluster is selected by at least one operation, so the mean of 2.21 distinct choices per cluster is the ceiling the cluster sizes allow, and the outputs that match do so because the operations compute the same quantity. Implementation identity therefore changes the selection under a fixed candidate set. This measures selection sensitivity and uses no reference method; it does not identify the better representative.

Pairwise, the median agreement is 52.4 per cent against a chance floor of 47.0, and 77 of the 210 pairs agree below chance (Figure 2). Collapsing the duplicate computations leaves eighteen distinct operations and 153 pairs, with a median of 50.0, so the spread is not an artefact of counting one computation twice. Among the source works read in full, none evaluates its operation against the others on common material.

Read against one fixed choice, the same swaps rewrite much of the stage output (Figure S4): of the twenty-one alternatives, five retain less than chance. **That comparison shares the clusters but not the frames**: the per-frame operations score one frame per second, while SIFT+BoVW runs its own key-frame selector.

LIQE was used for the complete-implementation substitution, having the largest generalization margin among the three eligible methods. The substitution changes 22 of the 42 representatives, the two operations agreeing in 20 clusters (47.6 per cent, 95% CI [32.4, 61.4] over the 18 source-aware blocks), close to the 47.0 per cent uniform-choice expectation. Whether the substitution also improves the output, measured with the technical branch of DOVER, is model-based evidence reported in the supplement, where for the complete substitution both intervals cross zero.

Under matched inputs the picture sharpens. Given at most ten uniformly spaced frames, 9.98 in common on average, and mean pooling for both, SIFT+BoVW and LIQE still select different representatives in 16 of the 42 clusters. Holding the scorer fixed while changing only sampling and pooling already moves 12 decisions against 22 for the complete operation; an exact McNemar test separates the two regimes ($p = .006$), as does the block sign-flip over the 9 topics ($p = .008$). Both scorer identity and input sampling carry part of the effect.

The retention percentages above put all 21 operations of Figure 2 on one frame per second, which makes their spread attributable to the scorer; Table S19 scores each complete operation under its own sampling.

4.7 Source-level synthesis of the audit

Table 5 brings together the roles the sources document and the results of the implementations evaluated here, organized by work and component. **The one audited source that names technical fidelity as the aim of a term**, Ref. [60], implements

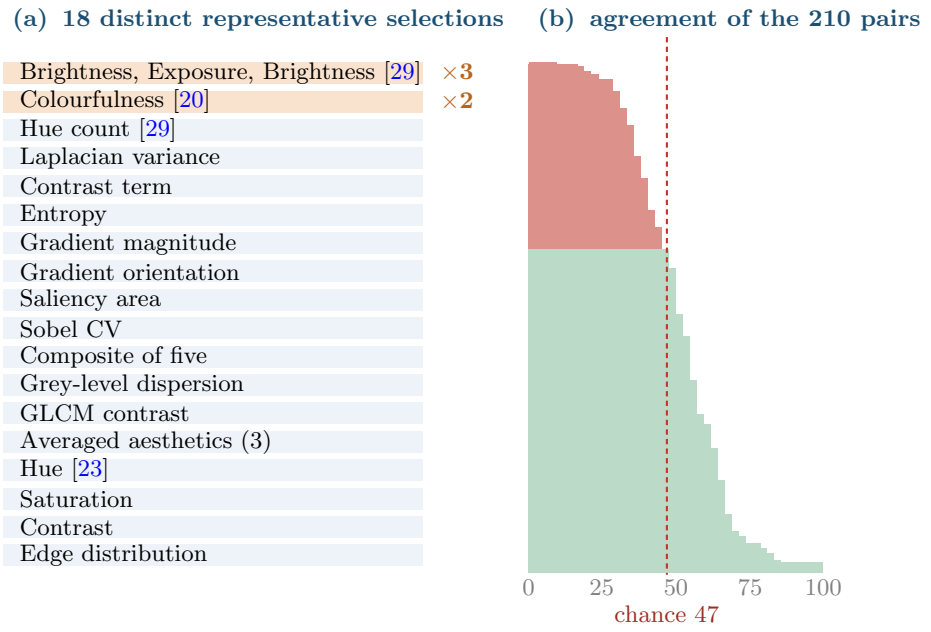


Fig. 2 The same 42 redundancy clusters and the same pipeline, with only the visual-selection operation swapped; **no reference model is involved**. **(a)**: the twenty-one operations grouped by the representative set each produces, giving eighteen *distinct sets*; a group of more than one operation carries its size as a multiplier. **(b)**: agreement for each of the 210 *unordered operation pairs*, sorted, against the chance floor of 47 per cent (dashed); **77 pairs agree below chance**. The only groups that agree completely are operations computing the same quantity: the brightness variants and the two colourfulness implementations. **SIFT+BoVW is absent from this panel by design**: every operation here scores the same input regime, and its key-frame selector is part of the published method, so holding the input fixed would evaluate a different operation. The matched-input control of Section 4.6 is the diagnostic that compares it under a common interface. *Labels*: CV, coefficient of variation; GLCM, grey-level co-occurrence matrix. Rows that name the same attribute carry the reference of the work that defines it.

it with BRISQUE, one of five dimensions of its regret-minimization query; evaluated alone here, BRISQUE does not meet both external conditions.

Three patterns emerge in the sources read in full: attribute borrowing, role reassignment, and summary-level validation. The counts are of confirmed uses, each with a verbatim excerpt.

Role reassignment shows the same computation carrying different claims. Laplacian variance appears as a threshold discarding blurred frames in Ref. [8] and as the dominant weighted term of a keyframe score in Ref. [30]. What was measured is the statistic *as a scorer*: it answers blur on every pair and inverts under noise. The published threshold was not run: a scorer can order a pair correctly while a threshold keeps or drops both members.

Attribute borrowing and the level of validation are read from each source. Of the eleven works whose full text is in the archive, two test a measure against human judgement of the property it names, evidence for a different construct: neither validates a single attribute as a stand-alone score of technical fidelity. By the provenance rule

Table 5 The audit read by source work. Each band is one source with the role it declares; each row is one component. **Degradation columns** give strict accuracy (per cent) on the constructed pairs, **50 being chance**, green where the operation prefers the original and red where it prefers the degraded copy; every cell prints its value, so the table reads in greyscale. *Dk* darkening, *Mo* motion blur, *Bl* Gaussian blur, *H2* H.264, *Rs* rescaling, *St* desaturation, *Ns* white noise, *Jp* JPEG. **Mean** is over all eight types, not the seven-type contrast of Section 4.4. **Agr** is chance-corrected agreement with LIQE, the designated reference for the literature-derived operations, over the 42 operational clusters. **Denominator**: 800 pairs (20 segments × 8 types × 5 levels), except the Ref. [30] band, which runs on the 11 segments carrying both of its inputs and therefore on 440 pairs. * the degradation acts directly on the quantity that operation measures; † the same computation is used by two source works, so the two rows are not independent evidence; ‡ the same Hasler expression, named by several sources; § a term of the published module implemented after the inventory freeze, outside the counts each band states; ¶ audited on the same footing but outside the eleven-work count, because it is the testbed's own operation. m the one model row: its agreement comes from the common-reference analysis of Section 4.5, which scores both families against LIQE. *n/a* marks a test that does not apply, and **a dash marks one that applies but was not run**.

	degradation type									
Source work / component	**Dk**	**Mo**	**Bl**	**H2**	**Rs**	**St**	**Ns**	**Jp**	**Mean**	**Agr**
Ref. [29] four features of a naive Bayes photo-quality classifier; *here: each feature alone, as a score;* **human ratings; per feature and combined**										
Edge distribution	70	*62**	*62**	11	74	97	100	31	**63.4**	−.124
Hue count	20	13	10	77	11	4	100	59	**36.8**	−.258
Contrast	*100**	100	100	98	100	30	10	60	**74.8**	+.326
Colourfulness‡	100	100	100	100	100	*100**	3	79	**85.2**	+.281
Ref. [51] average five attributes into a per-frame quality score; *here: its mean rule, on our attribute vector;* summary-level evaluation										
The mean rule, our vector	100	100	100	100	100	98	29	69	**87.0**	+.371
Ref. [44] four audited terms across three objectives, and a five-term attention module; *here: the published rule over the vector implemented here, then each term;* summary-level evaluation										
Visual-attention module	100	100	100	100	100	100	0	20	**77.5**	n/a
Colourfulness‡	100	100	100	100	100	*100**	3	79	**85.2**	+.281
Gradient magnitude	100	100	100	100	100	23	0	5	**66.0**	+.551
Saliency	62	29	20	94	21	18	13	10	**33.4**	+.101
Contrast term§	0	0	0	3	0	68	81	55	**25.9**	—
Motion magnitude§	100	87	100	98	98	14	9	10	**64.5**	—
Entropy	100	77	83	10	77	97	0	66	**63.8**	+.236
Gradient orientation	100	80	26	95	9	21	0	100	**53.9**	+.461
Ref. [30] .7 *z*(sharp) + .3 *z*(bright) as a keyframe score, **on 11 of the 20 segments**; *here: the rule, our normalization population;* **no isolated test of the rule**										
The published rule	100	100	100	100	100	0	0	49	**68.6**	n/a
Sharpness alone, same 11†	100	100	100	91	100	47	0	49	**73.4**	n/a
Ref. [8] a threshold discarding blurred frames, not a score; *here: the statistic as a scorer, not the filter;* summary-level evaluation, with ablation										
Laplacian variance†	100	*100**	*100**	95	100	50	0	66	**76.4**	+.506
Ref. [27] a badness term in a cost multiplicative in cluster distance; *here: the term, read increasing;* summary-level evaluation, with an optimisation ablation										
Grey-level dispersion†	*100**	100	100	100	100	22	25	90	**79.6**	+.191
Ref. [60] a discard filter for meaningless frames, and BRISQUE as one of five score dimensions; *here: the filter statistic and the fidelity dimension, read alone;* summary-level evaluation, with ablation										
Grey-level dispersion†	*100**	100	100	100	100	22	25	90	**79.6**	+.191
BRISQUE, the fidelity dimension m	100	100	100	99	100	0	67	80	**80.8**	+.685
Ref. [5]¶ a TF-IDF sum over a visual-word vocabulary, scoring whether a segment holds images of interest (a relevance criterion); *here: the score as the source specifies it; the testbed's own operation*										
SIFT+BoVW sum¶	100	100	90	90	88	39	8	6	**65.1**	+.011
Ref. [61] one texture term of several; *here: the term alone;* summary-level evaluation, with ablation										
GLCM contrast	100	100	100	100	100	67	0	72	**79.9**	+.461
Ref. [23] one aesthetic attribute of several; *here: the attribute alone;* summary-level evaluation										
Saturation	13	100	100	15	100	*100**	1	90	**64.9**	+.011
Ref. [19] three aesthetic attributes, listed without a combination; *here: the raw mean;* summary-level evaluation, with ablation										
Averaged aesthetics	100	100	100	100	100	100	3	67	**83.8**	+.461
Ref. [20] **the one autonomous index**, and it indexes colourfulness; *here: the index as published;* **human ratings of the named property**										
Colourfulness‡	100	100	100	100	100	*100**	3	79	**85.2**	+.281

of Section 3.1, one of the sixteen operations is a stand-alone index in its source, and it indexes colourfulness rather than quality; the rest enter as classifier features, terms of a sum, or signals in a discard filter. Among the remaining works, eight report summary-level evaluation, which does not isolate the component's validity, and the one left reports no isolated test of its own rule or weights, so the three categories account for all eleven.

5 Discussion and limitations

The central finding of this study is that visual-selection statistics derived from the video summarization literature, when reused as standalone technical-quality scores, fail to satisfy the technical-fidelity requirements proposed here. The algebraic factorization of the SIFT+BoVW raw-count score demonstrates that its behavior is overwhelmingly driven by descriptor count rather than visual fidelity; replacing the score with the raw descriptor count preserves every single choice across the 42 operational clusters. For other operations, the concentration of reversals under white noise is consistent with noise inflating local high-frequency detail, which these statistics can misread as superior quality. At the application level, our evaluation separates scoring behavior from output preference, establishing selection sensitivity rather than guaranteeing product improvement. Consequently, practical reuses of these operations require explicit documentation of the intended property, exact statistics, declared polarity, sampling, and aggregation regimes. Although dedicated no-reference image quality assessment (NR-IQA) models offer better-supported alternatives, their robustness remains tightly bound to training exposure, distortion types, and domain characteristics. These findings reframe two readings in the current literature: first, extending the evaluation-side insight of Ref. [40], that protocol choices can dominate summarization scores, to the selection stage; and second, showing that isolated visual attributes frequently fail when extracted from the broader objective functions in which they were originally embedded.

These conclusions must be interpreted alongside some inherent limitations. First, while the quantitative measurements are fully deterministic given the benchmark inventory, the upstream process of compiling the archive, coding each source's functional role, and recording validation evidence relied on manual analysis by a single author. To ensure reproducibility and verification, all eleven source works were read in full, and every attribution is anchored to recorded verbatim excerpts (Tables S24 and S25). Furthermore, our experimental scope is restricted to selected implementations across three image-quality databases and one broadcast-news pipeline; as a result, these conclusions do not automatically extend to different tasks, model families, or newer neural checkpoints.

The design of the benchmark and inference protocols also introduces specific constraints. The constructed evaluation benchmark comprises twenty video segments organized into 12 source-aware blocks, where ground-truth fidelity is established via controlled algorithmic transformations rather than subjective human preference. Statistical inference remains sensitive to small-sample tail conventions and spatial dependence assumptions, particularly given that eligibility thresholds and most inference parameters were not preregistered. Consequently, analytical variations such as

best-component contrasts, oracle polarity alignment, and in-sample ridge regressions serve purely as diagnostic insights rather than prospective validations.

At the operational application level, the 42 redundancy clusters represent operationally redundant segment alternatives nested across 19 news topics with shared source material, rather than strictly semantically equivalent recordings. Because no human subjective quality ratings were collected for these specific segments, the reference benchmarks evaluate response to controlled signal degradation rather than human preference between distinct video captures. Technical quality here functions strictly as a tie-breaker within redundancy groups rather than a holistic verdict on the resulting summary. Similarly, external DOVER comparisons cannot isolate temporal modeling effects from underlying neural architecture or training data, and the official standardized fusion model was omitted.

Finally, the auxiliary diagnostic analyses presented are descriptive rather than confirmatory. In particular, conditioning the SIFT+BoVW score directly on keypoint counts risks removing legitimate quality signals alongside the dominant count channel. While the archived execution outputs allow full reconstruction of all derived statistics, the original software execution environment is not bit-identical. Lastly, partial-coverage models and later deep control architectures remain outside the primary screening comparisons evaluated in this audit.

6 Conclusion

Visual-selection operations from the summarization literature, reused as standalone technical-quality scores, fail in that role, and the choice among them changes what a pipeline selects. Every one of the fifteen directionally specified operations prefers the degraded copy on at least one controlled transformation, nine keep that reversal under the most conservative analysis, and none of them, nor any of the nineteen attribute configurations, meets both external screening conditions; three of ten dedicated NR-IQA models do. At 74.1 per cent the deep proxy sits below eight of the fifteen literature operations and below every dedicated model; most of that survives when the cluster members come from other topics, so what it detects is a copy made atypical by degradation, which is representativeness standing in for quality.

The failures share a pattern. Under white noise, the one transformation that adds local high-frequency content, thirteen of the fifteen operations reverse, and for the one audited score whose terms are recoverable, the SIFT+BoVW sum of Ref. [5], an exact factorization shows why: descriptor count carries the sign of the score difference in 797 of 800 pairs, and replacing the score by that count reproduces all 42 representative choices. The operation reads the amount of local detail as quality; the mechanism is measured for that score and consistent with the noise reversals of the others. The choice among these operations is consequential: on 42 fixed clusters, 21 implementations yield 18 distinct selections.

Three studies remain: a blind pairwise judgement of real candidate recordings, which would put the technical-quality ordering and a viewer's on the same pairs; preference between whole summaries, with summaries as the unit; and protocols for

context-dependent selection rules. The results are bounded by the evaluated configurations, three image-quality databases and one pipeline. A reused statistic should be validated against the property and decision role it is intended to serve.

Statements and Declarations

Data availability

KADID-10k, KonIQ-10k and LIVE In the Wild are public. The supplementary material (Online Resource 1, containing Supplementary Tables S1–S33 and Supplementary Figures S1–S4) and the reproducibility archive (Online Resource 2) are available at https://github.com/Prism411/evaluality-mtap-submission; scripts, derived scores, clusters, seeds and latency results are in that archive, and a Git bundle that makes the registered-prediction commit independently verifiable will accompany the permanent release. **The broadcast corpus itself is not redistributed**: it was assembled by colleagues at our laboratory and remains with its authors, and this study uses it under their permission rather than publishing it. What the archive contains for that corpus is derived material, namely per-segment scores, cluster memberships, seeds and timings, together with a two-topic, six-video sample so that the scoring scripts can be run on real input; owing to size, the archive released for review carries one of the two topics.

Code availability

Analysis scripts and machine-readable results are in the same repository (Online Resource 2); a release tagged to the accepted version will be archived at publication.

Competing interests

The authors declare no competing financial interests. Two non-financial relationships are disclosed: the modular pipeline used as testbed and the broadcast corpus it runs on were developed at the authors' laboratory, as stated in Section 3.2, and R. G. is a co-author of [6], one of the works audited here.

Author contributions

J. L. S. G. designed and implemented the evaluation protocol, ran the experiments and wrote the manuscript. R. G. supervised the research, defined the problem framing and revised the manuscript. Both authors approved the final version.

Ethics approval, consent to participate and consent for publication

Not applicable. The study involved no recruitment, intervention or new data collection from human participants.

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Use of generative artificial intelligence

Generative AI tools (OpenAI Codex and Anthropic Claude) were used to assist in writing the measurement and analysis scripts. All of their output was reviewed, tested and verified by the authors, who designed the study, interpreted the results and take full responsibility for the content of this article.

References

[1] Agnolucci L, Galteri L, Bertini M (2024) Quality-aware image-text alignment for opinion-unaware image quality assessment. arXiv preprint https://doi.org/10.48550/arXiv.2403.11176, URL https://arxiv.org/abs/2403.11176, version 3, March 2025

[2] Agnolucci L, Galteri L, Bertini M, et al (2024) ARNIQA: Learning distortion manifold for image quality assessment. In: Proc. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp 189–198, https://doi.org/10.1109/WACV57701.2024.00026

[3] Apostolidis E, Adamantidou E, Metsai AI, et al (2021) Video summarization using deep neural networks: A survey. Proceedings of the IEEE 109(11):1838–1863. https://doi.org/10.1109/JPROC.2021.3117472

[4] Bano S, Cavallaro A (2016) Vicomp: composition of user-generated videos. Multimedia Tools and Applications 75(12):7187–7210. https://doi.org/10.1007/s11042-015-2641-2

[5] Barbieri TTdS (2021) Sumarização automática multivídeo baseada em estratégias humanas. PhD thesis, Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, https://doi.org/10.11606/T.55.2021.tde-27092021-101106

[6] Barbieri TTdS, Goularte R (2021) Content selection criteria for news multi-video summarization based on human strategies. International Journal on Digital Libraries 22:1–14. https://doi.org/10.1007/s00799-020-00281-9

[7] Behzadpour M, Ghanbari M (2023) Improving precision of objective image/video quality meters. Multimedia Tools and Applications 82(3):4465–4478. https://doi.org/10.1007/s11042-022-13416-8

[8] Cakmak MC, Agarwal N, Poudel D (2025) TriPSS: A tri-modal keyframe extraction framework using perceptual, structural and semantic representations. ArXiv:2506.05395v2

[9] Cao P, Li D, Ma K (2024) Image quality assessment: Integrating model-centric and data-centric approaches. In: Proc. Conference on Parsimony and Learning (CPAL), pp 529–541, arXiv:2207.14769

[10] Chen C, Mo J, Hou J, et al (2024) TOPIQ: A top-down approach from semantics to distortions for image quality assessment. IEEE Transactions on Image Processing 33:2404–2418. https://doi.org/10.1109/TIP.2024.3378466

[11] Cirne MVM, Pedrini H (2014) Summarization of videos by image quality assessment. In: Progress in Pattern Recognition, Image Analysis, Computer Vision, and Applications, Lecture Notes in Computer Science, vol 8827. Springer, pp 901–908, https://doi.org/10.1007/978-3-319-12568-8_109

[12] Cricri F, Dabov K, Curcio IDD, et al (2014) Multimodal extraction of events and of information about the recording activity in user generated videos. Multimedia Tools and Applications 70(1):119–158. https://doi.org/10.1007/s11042-012-1085-1

[13] Cronbach LJ, Meehl PE (1955) Construct validity in psychological tests. Psychological Bulletin 52(4):281–302. https://doi.org/10.1037/h0040957

[14] Csurka G, Dance CR, Fan L, et al (2004) Visual categorization with bags of keypoints. In: Workshop on Statistical Learning in Computer Vision, ECCV, pp 1–22

[15] Darcet T, Oquab M, Mairal J, et al (2024) Vision transformers need registers. In: Proc. ICLR, oral. arXiv:2309.16588

[16] Ghadiyaram D, Bovik AC (2016) Massive online crowdsourced study of subjective and objective picture quality. IEEE Transactions on Image Processing 25(1):372–387. https://doi.org/10.1109/TIP.2015.2500021

[17] Golestaneh SA, Dadsetan S, Kitani KM (2022) No-reference image quality assessment via transformers, relative ranking, and self-consistency. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp 1220–1230

[18] Gonçalves Moraes L (2024) HSMVideoSumm: open-video-summary. https://github.com/leogoncalvesm/open-video-summary, implementation of the multi-video summarization pipeline

[19] Gygli M, Grabner H, Riemenschneider H, et al (2014) Creating summaries from user videos. In: ECCV, pp 505–520, https://doi.org/10.1007/978-3-319-10584-0_33

[20] Hasler D, Süsstrunk S (2003) Measuring colourfulness in natural images. In: Human Vision and Electronic Imaging VIII, pp 87–95, https://doi.org/10.1117/12.477378

[21] Hosu V, Lin H, Sziranyi T, et al (2020) KonIQ-10k: An ecologically valid database for deep learning of blind image quality assessment. IEEE Transactions on Image

Processing 29:4041–4056. https://doi.org/10.1109/TIP.2020.2967829

[22] Hou X, Zhang L (2007) Saliency detection: A spectral residual approach. In: 2007 IEEE Conference on Computer Vision and Pattern Recognition, pp 1–8, https://doi.org/10.1109/CVPR.2007.383267

[23] Huang H, Wu Z, Pang G, et al (2024) An aesthetic-driven approach to unsupervised video summarization. IEEE Access 12. https://doi.org/10.1109/ACCESS.2024.3434508

[24] Huang X, Li C, Bentaleb A, et al (2023) XGC-VQA: A unified video quality assessment model for user, professionally, and occupationally-generated content. In: Proc. IEEE ICMEW, https://doi.org/10.1109/ICMEW59549.2023.00081

[25] International Telecommunication Union (2008) Subjective video quality assessment methods for multimedia applications. ITU-T Recommendation P.910

[26] Isogawa M, Mikami D, Takahashi K, et al (2019) Image quality assessment for inpainted images via learning to rank. Multimedia Tools and Applications 78(2):1399–1418. https://doi.org/10.1007/s11042-018-6186-z

[27] Jeong Dj, Yoo HJ, Cho NI (2017) A static video summarization method based on the sparse coding of features and representativeness of frames. EURASIP Journal on Image and Video Processing 2017(1):1. https://doi.org/10.1186/s13640-016-0122-9

[28] Ke J, Wang Q, Wang Y, et al (2021) MUSIQ: Multi-scale image quality transformer. In: Proc. IEEE/CVF International Conference on Computer Vision (ICCV), pp 5148–5157, https://doi.org/10.1109/ICCV48922.2021.00510

[29] Ke Y, Tang X, Jing F (2006) The design of high-level features for photo quality assessment. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 419–426, https://doi.org/10.1109/CVPR.2006.303

[30] Korolkov V (2025) Scene detection policies and keyframe extraction strategies for large-scale video analysis. ArXiv:2506.00667

[31] Krasula L, Fliegel K, Le Callet P, et al (2016) On the accuracy of objective image and video quality models: New methodology for performance evaluation. In: Proc. 8th International Conference on Quality of Multimedia Experience (QoMEX), https://doi.org/10.1109/QoMEX.2016.7498936

[32] Leszczuk M (2014) Optimising task-based video quality: A journey from subjective psychophysical experiments to objective quality optimisation. Multimedia Tools and Applications 68(1):41–58. https://doi.org/10.1007/s11042-012-1161-6

[33] Li Y, Merialdo B (2010) Multi-video summarization based on Video-MMR. In: Proc. WIAMIS

[34] Lin H, Hosu V, Saupe D (2019) KADID-10k: A large-scale artificially distorted IQA database. In: Proc. QoMEX, pp 1–3, https://doi.org/10.1109/QoMEX.2019.8743252

[35] Lowe DG (2004) Distinctive image features from scale-invariant keypoints. International Journal of Computer Vision 60(2):91–110. https://doi.org/10.1023/B:VISI.0000029664.99615.94

[36] Mittal A, Moorthy AK, Bovik AC (2012) No-reference image quality assessment in the spatial domain. IEEE Transactions on Image Processing 21(12):4695–4708. https://doi.org/10.1109/TIP.2012.2214050

[37] Mittal A, Soundararajan R, Bovik AC (2013) Making a "completely blind" image quality analyzer. IEEE Signal Processing Letters 20(3):209–212. https://doi.org/10.1109/LSP.2012.2227726

[38] Oquab M, Darcet T, Moutakanni T, et al (2024) DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research (TMLR) ArXiv:2304.07193

[39] Oszust M (2017) Image quality assessment with lasso regression and pairwise score differences. Multimedia Tools and Applications 76(11):13255–13270. https://doi.org/10.1007/s11042-016-3755-x

[40] Otani M, Nakashima Y, Rahtu E, et al (2019) Rethinking the evaluation of video summaries. In: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 7588–7596, https://doi.org/10.1109/CVPR.2019.00778

[41] Panda R, Roy-Chowdhury AK (2017) Collaborative summarization of topic-related videos. In: Proc. CVPR, pp 4274–4283, https://doi.org/10.1109/CVPR.2017.455

[42] Panda R, Mithun NC, Roy-Chowdhury AK (2017) Diversity-aware multi-video summarization. IEEE Transactions on Image Processing 26(10):4712–4724. https://doi.org/10.1109/TIP.2017.2708902

[43] Pinson MH (2023) Why no reference metrics for image and video quality lack accuracy and reproducibility. IEEE Transactions on Broadcasting 69(1):97–117. https://doi.org/10.1109/TBC.2022.3191059

[44] Qaroush AM, Jubran M, Olayyan Q, et al (2025) Maximum relevant diversity aware, multi-video summarization using clustering and evolutionally multi-objective optimization. Expert Systems with Applications 292:128631. https://doi.org/10.1016/j.eswa.2025.128631

[45] Radford A, Kim JW, Hallacy C, et al (2021) Learning transferable visual models from natural language supervision. In: Proc. ICML, pp 8748–8763, pMLR 139

[46] Robitza W, Rao RR, Göring S, et al (2021) Impact of spatial and temporal information on video quality and compressibility. In: Proc. IEEE QoMEX, pp 65–70, https://doi.org/10.1109/QoMEX51781.2021.9465452

[47] Salton G, Buckley C (1988) Term-weighting approaches in automatic text retrieval. Information Processing & Management 24(5):513–523. https://doi.org/10.1016/0306-4573(88)90021-0

[48] Sheikh HR, Sabir MF, Bovik AC (2006) A statistical evaluation of recent full reference image quality assessment algorithms. IEEE Transactions on Image Processing 15(11):3440–3451. https://doi.org/10.1109/TIP.2006.881959

[49] Sivic J, Zisserman A (2003) Video Google: A text retrieval approach to object matching in videos. In: Proc. ICCV, pp 1470–1477, https://doi.org/10.1109/ICCV.2003.1238663

[50] Song Y, Vallmitjana J, Stent A, et al (2015) TVSum: Summarizing web videos using titles. In: Proc. CVPR, pp 5179–5187, https://doi.org/10.1109/CVPR.2015.7299154

[51] Srinivas M, Pai MMM, Pai RM (2016) An improved algorithm for video summarization — a rank based approach. Procedia Computer Science 89:812–819. https://doi.org/10.1016/j.procs.2016.06.065

[52] Su S, Yan Q, Zhu Y, et al (2020) Blindly assess image quality in the wild guided by a self-adaptive hyper network. In: Proc. CVPR, pp 3667–3676, https://doi.org/10.1109/CVPR42600.2020.00372

[53] Truong BT, Venkatesh S (2007) Video abstraction: A systematic review and classification. ACM Transactions on Multimedia Computing, Communications, and Applications 3(1):1–37. https://doi.org/10.1145/1198302.1198305

[54] Wang J, Chan KCK, Loy CC (2023) Exploring CLIP for assessing the look and feel of images. In: Proc. AAAI, pp 2555–2563, https://doi.org/10.1609/aaai.v37i2.25353

[55] Wang Y, Ke J, Talebi H, et al (2021) Rich features for perceptual quality assessment of UGC videos. In: Proc. CVPR, pp 13435–13444, https://doi.org/10.1109/CVPR46437.2021.01323

[56] Wang Z, Bovik AC, Sheikh HR, et al (2004) Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing 13(4):600–612. https://doi.org/10.1109/TIP.2003.819861

[57] Wu H, Zhang E, Liao L, et al (2023) Exploring video quality assessment on user generated contents from aesthetic and technical perspectives. In: Proc. ICCV, pp 20144–20154, https://doi.org/10.1109/ICCV51070.2023.01843

[58] Wu H, Zhang Z, Zhang W, et al (2024) Q-Align: Teaching LMMs for visual scoring via discrete text-defined levels. In: Proc. ICML, pp 54015–54029, pMLR 235, arXiv:2312.17090

[59] Wu J, Zhong Sh, Liu Y (2019) MvsGCN: A novel graph convolutional network for multi-video summarization. In: Proc. ACM MM, pp 2254–2262, https://doi.org/10.1145/3343031.3350938

[60] Xu Y, Zheng J, Tao Y (2024) Property constrained video summarization via regret minimization. SN Computer Science 5:254. https://doi.org/10.1007/s42979-023-02588-1

[61] Yang G, He Z, Su Z, et al (2024) Keyframe recommendation based on feature intercross and fusion. Complex & Intelligent Systems 10:4955–4971. https://doi.org/10.1007/s40747-024-01417-z

[62] Yang S, Wu T, Shi S, et al (2022) MANIQA: Multi-dimension attention network for no-reference image quality assessment. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, pp 1191–1200

[63] Zhai G, Min X (2020) Perceptual image quality assessment: A survey. Science China Information Sciences 63(11):211301. https://doi.org/10.1007/s11432-019-2757-1

[64] Zhang W, Ma K, Yan J, et al (2020) Blind image quality assessment using a deep bilinear convolutional neural network. IEEE TCSVT 30(1):36–47. https://doi.org/10.1109/TCSVT.2018.2886771

[65] Zhang W, Zhai G, Wei Y, et al (2023) Blind image quality assessment via vision-language correspondence: A multitask learning perspective. In: Proc. CVPR, pp 14071–14081, https://doi.org/10.1109/CVPR52729.2023.01352